

Test Cases

From assignment:

Enter a string (or 'END' to quit): connecting to 192.168.1.1 now

Extracted IPv4 address: 192.168.1.1 (decimal value: 3232235777, port: none)

Enter a string (or 'END' to quit): server=10.0.0.255:8080end

Extracted IPv4 address: 10.0.0.255 (decimal value: 167772415, port: 8080)

Enter a string (or 'END' to quit): 192a168.1.1.1

Extracted IPv4 address: 168.1.1.1 (decimal value: 2818638081, port: none)

Enter a string (or 'END' to quit): 192.168.1.1.

Invalid input: no valid IPv4 address found

Enter a string (or 'END' to quit): Connection from 192.168.1.1 refused

Extracted IPv4 address: 192.168.1.1 (decimal value: 3232235777, port: none)

Enter a string (or 'END' to quit): 192.168.01.1

Invalid input: no valid IPv4 address found

Enter a string (or 'END' to quit): 1.2.3.4:99999

Invalid input: no valid IPv4 address found

Enter a string (or 'END' to quit): 12.34.56

Invalid input: no valid IPv4 address found

Enter a string (or 'END' to quit): no number here

Invalid input: no valid IPv4 address found

Enter a string (or 'END' to quit): END

Program terminated.

Additional cases:

Enter a string (or 'END' to quit): 256.256.256.256

Invalid input: no valid IPv4 address found

Enter a string (or 'END' to quit): 256.256.256.256:500

Invalid input: no valid IPv4 address found

Enter a string (or 'END' to quit): 192.168.1.170:

Invalid input: no valid IPv4 address found

Enter a string (or 'END' to quit): 192.168.1.170:0

Extracted IPv4 address: 192.168.1.170 (decimal value: 3232235946, port: 0)

Enter a string (or 'END' to quit): 192.168.1.170:99999

Invalid input: no valid IPv4 address found

Enter a string (or 'END' to quit): 192.168.1.170:65535

Extracted IPv4 address: 192.168.1.170 (decimal value: 3232235946, port: 65535)

Enter a string (or 'END' to quit): 192.168.1.170:65536

Invalid input: no valid IPv4 address found

Enter a string (or 'END' to quit): 0.0.0.0

Extracted IPv4 address: 0.0.0.0 (decimal value: 0, port: none)

Enter a string (or 'END' to quit): 0.0.0.0:01

Invalid input: no valid IPv4 address found

Enter a string (or 'END' to quit): 0.0.0.0:1

Extracted IPv4 address: 0.0.0.0 (decimal value: 0, port: 1)

Enter a string (or 'END' to quit): 192.168.1.170sdfasdf:500

Extracted IPv4 address: 192.168.1.170 (decimal value: 3232235946, port: none)

Enter a string (or 'END' to quit): 500:192.168:1:170

Invalid input: no valid IPv4 address found

Enter a string (or 'END' to quit): :500

Invalid input: no valid IPv4 address found

Enter a string (or 'END' to quit): 192.168:1:170:-100

Invalid input: no valid IPv4 address found

Enter a string (or 'END' to quit): 0.0.0.256

Invalid input: no valid IPv4 address found

Enter a string (or 'END' to quit): hello192.168.1.1

Extracted IPv4 address: 192.168.1.1 (decimal value: 3232235777, port: none)

Enter a string (or 'END' to quit): Could not find address 192.168.2.5

Extracted IPv4 address: 192.168.2.5 (decimal value: 3232236037, port: none)

Enter a string (or 'END' to quit): 0.0.0.0:0

Extracted IPv4 address: 0.0.0.0 (decimal value: 0, port: 0)

Enter a string (or 'END' to quit): 192.168.1.20 END

Extracted IPv4 address: 192.168.1.20 (decimal value: 3232235796, port: none)

Enter a string (or 'END' to quit): END 192.168.2.0

Extracted IPv4 address: 192.168.2.0 (decimal value: 3232236032, port: none)

Enter a string (or 'END' to quit): 192.168.1.25:00000

Invalid input: no valid IPv4 address found

Enter a string (or 'END' to quit): 192.168.1.25:10000

Extracted IPv4 address: 192.168.1.25 (decimal value: 3232235801, port: 10000)

Enter a string (or 'END' to quit): 192.168.1.25:000000

Invalid input: no valid IPv4 address found

Enter a string (or 'END' to quit):